



Novel Uncertainty Principles for Two-Sided Quaternion Linear Canonical Transform

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Abstract. The uncertainty principle, which offers information about a function and its Fourier transform in the time-frequency plane, is particularly powerful in mathematics, physics and signal processing community. In this paper, based on the fundamental relationship between the quaternion linear canonical transform (QLCT) and quaternion Fourier transform (QFT), we propose two different uncertainty principles for the two-sided QLCT. It is shown that the lower bounds can be obtained on the product of spreads of a quaternion-valued function and its two-sided QLCT from newly derived results. Furthermore, an example is given to verify the consequences. Finally, some possible applications are provided to demonstrate the usefulness of new uncertainty relations in the QLCT domain.

Keywords. Uncertainty principle, Quaternion linear canonical transform, Quaternion Fourier transform.

1. Introduction

The uncertainty principle (UP), which provides a lower bound on the spreads of two specific transform domains, is of importance in various scientific fields such as mathematics, signal processing and information theory [9, 12, 16]. In quantum mechanics, one UP demonstrates that the impossibility of simultaneous precise measurements of a particle's momentum and its position. In signal processing, it states that a signal cannot be simultaneously sharply located in both time and frequency domains. In mathematics, the classical Heisenberg UP in the time-frequency plane is given by [29, 31]

$$\int_{\mathbb{R}} (t - t_0)^2 |f(t)|^2 dt \cdot \int_{\mathbb{R}} (w - w_0)^2 |F(w)|^2 dw > \frac{1}{4} \left(\int_{\mathbb{R}} |f(t)|^2 dt \right)^2 \quad (1)$$

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where

$$\begin{aligned} t_0 &= \frac{1}{E} \int_{\mathbb{R}} t |f(t)|^2 dt \\ w_0 &= \frac{1}{E} \int_{\mathbb{R}} w |F(w)|^2 dw \end{aligned} \quad (2)$$

Herein, $E = \int_{\mathbb{R}} |f(t)|^2 dt = \int_{\mathbb{R}} |F(w)|^2 dw$ denotes the energy of $f(t)$. $F(w)$ denotes the Fourier transform (FT) of the signal $f(t)$. t_0 and w_0 stand for the mean time and mean frequency, respectively. We often let $t_0 = w_0 = 0$ for simplicity because UPs do not depend on the location of means [36, 40]. The UP in Eq. (1) describes that the product of spreads of a given signal $f(t)$ in time and frequency domains is limited by a lower bound. In terms of the importance of the UP in mathematics, physics, optics and signal processing, there are many efforts to extend it into various types of functions and integral transforms in one dimensional condition, see [11, 19, 27, 29, 31, 32, 35–38, 42, 43].

Recently, it has become popular to generalize the integral transforms from real and complex numbers to quaternion algebra to study higher dimension such as the quaternion Fourier transform (QFT) [17], the fractional quaternion Fourier transform (FRQFT) [33, 34], the quaternion wavelet transform (QWT) [7], the quaternion linear canonical transform (QLCT) [22, 23] and others [2, 14]. The quaternion algebra was first invented by W. R. Hamilton in 1843 and is denoted by $\mathbb{H}$ in his honor. Due to the non-commutative property of quaternion multiplication, there are three different types of two-dimension (2D) quaternion integral transform. For example, there are three types of the QFT, which are the left-sided QFT, the right-sided QFT and the two-sided QFT. Many properties of the classical FT have been extended for quaternion types including shift, modulation, differentiation, energy conservation, convolution and correlation, UP and so on. The conventional UP in Eq. (1) has been generalized for the right-sided QFT firstly by Bahri [6] in the name of component-wise UP, which takes the form of

$$\int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} u_k^2 |\mathcal{F}_r \{f\}(\mathbf{u})|^2 d\mathbf{u} \geq \frac{1}{4} \left(\int_{\mathbb{R}^2} |f(\mathbf{x})|^2 d\mathbf{x} \right)^2, \quad k = 1, 2 \quad (3)$$

where $\mathbf{x}, \mathbf{u} \in \mathbb{R}^2$, $\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$, $\mathbf{u} = u_1 \mathbf{e}_1 + u_2 \mathbf{e}_2$. $\mathbf{e}_1$ and $\mathbf{e}_2$ are the unit 2D vectors and orthogonal to each other (That is, the scalar dot product is zero, $\mathbf{e}_1 \cdot \mathbf{e}_2 = 0$). $\mathcal{F}_r$ is the right-sided QFT operator. The UP in Eq. (3) provides a lower bound on the product of the effective widths of quaternion-valued functions in spatial and frequency domains, which is also valid for the left-sided QFT and two-sided QFT. The work of Bahri motivates a large amount of attempts on generalizing various UPs from complex FT into the QFT setting. To be specific, Hitzer proposed a directional uncertainty principle associated with the two-sided QFT by quaternion split, which is given as follows [21]

$$\begin{aligned}
& \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} (\boldsymbol{\beta} \cdot \mathbf{u})^2 |\mathcal{F}\{f\}(\mathbf{u})|^2 d\mathbf{u} \\
& \geq \frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \boldsymbol{\beta})^2 F_-^2 + (\boldsymbol{\alpha} \cdot \boldsymbol{\beta}')^2 F_+^2 \right]
\end{aligned} \tag{4}$$

where $F_- = \int_{\mathbb{R}^2} |f_-(\mathbf{x})|^2 d\mathbf{x}$, $F_+ = \int_{\mathbb{R}^2} |f_+(\mathbf{x})|^2 d\mathbf{x}$, constant vectors $\boldsymbol{\alpha}, \boldsymbol{\beta} \in \mathbb{R}^2$ and $\boldsymbol{\beta}' = -\beta_1 \mathbf{e}_1 + \beta_2 \mathbf{e}_2$. f_+ and f_- are two parts of any quaternion function f detailed in Sect. 2. It is worth noting that the UP in Eq. (4) is a generalization of Eq. (3) when choosing $\boldsymbol{\alpha} = \boldsymbol{\beta} = \mathbf{e}_\mathbf{x}$. In order to obtain a better lower bound of UP to the QFT, Bahri formulated a modified uncertainty principle for the two-sided QFT in [4], which is rewritten as following

$$\begin{aligned}
& \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \int_{\mathbb{R}^2} u_1^2 |\mathcal{F}(\mathbf{u})|^2 u_2^2 d\mathbf{u} \\
& \geq \left| \frac{1}{2} F - \int_{\mathbb{R}^2} \left| x_1 x_2 \frac{\partial f(\mathbf{x})}{\partial x_2} \frac{\partial f(\mathbf{x})}{\partial x_1} \right| d\mathbf{x} \right|^2
\end{aligned} \tag{5}$$

where $F = \int_{\mathbb{R}^2} |f(\mathbf{x})|^2 d\mathbf{x}$ denotes the signal energy and $\bar{f}$ means conjugate of f . Other valuable and interesting work on different forms of UPs in the QFT domain can refer to [8, 40, 41]. With the rapid development of UPs with the QFT in mathematics, optics and physics, it is worthwhile and valuable to investigate it under other significant integral transforms as well.

The linear canonical transform (LCT) was first proposed by Moshinsky and Collins in the 1970s [10, 28]. It is a three free parameter class of linear integral transformation and includes many important integral transforms as its special cases such as the FT, the fractional Fourier transform (FRFT), the Fresnel transform (FST), the Lorentz transform (LT), and the other transforms. The LCT is an effective tool for chirp signals and used in many fields of optics and signal processing [15, 20]. As shown in [22–24], the LCT has been investigated with the context of quaternion algebra, and plays a vital role in the representation of hyper-complex signal. Since the UP has been extended to the QFT and played a vital role in quaternion algebra et al., it is natural to ask what kind of the UP in spatial and LCT frequency domains obeys? The component-wise UPs in Eq. (3) for the QLCT have been derived by Kou [23, 24], Fu [18] and Bahri [5] through different methods. The directional UP in Eq. (4) is a directional case for quaternion valued functions subject to the QFT. Yang and Kou [39] further investigated a unified UP for the spatial and frequency domains. Specifically, a directional case for complex paravector-valued signals is proposed by them under the frame of Clifford algebra. Clifford algebra is a generalization of the real numbers, complex numbers quaternions and several other hyper-complex number systems. Other type of the UP related to the QLCT, for example, the logarithmic UP, has been established recently [1]. To the best of our knowledge, the modified UP in Eq. (5) and the directional UP in Eq. (4) for the QLCT with respect to arbitrary directions have not derived yet. It is therefore interesting and worthwhile to investigate these kinds of UPs and obtain more valuable results associated with the QLCT.

The main objective of this paper is to establish two kinds of UPs for the QLCT. The rest of paper is organized as follows. Sect. 2 provides a brief introduction to some general definitions on quaternion algebra and the QLCT. The relationship between the QFT and QLCT is further reviewed in this section. The main results are presented in Sect. 3. Theorems 1 and 2 are the directional UP and modified UP for the QLCT, respectively. To further understand the derived consequences, an example is given which achieves the lower bound of Theorem 2. Some potential applications are provided to discuss the usefulness of new uncertainty relations in the QLCT domain in Sect. 4. Finally, Sect. 5 concludes this paper and points out its future research directions.

2. Preliminaries

2.1. Quaternion Algebra

The quaternion algebra [26] was first invented by Hamilton in 1843 and is denoted by $\mathbb{H}$ in his honor. Every element of $\mathbb{H}$ can be written in the following form

$$\mathbb{H} = \{q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} | q_0, q_1, q_2, q_3 \in \mathbb{R}\} \quad (6)$$

where $\mathbf{i}, \mathbf{j}$ and $\mathbf{k}$ are three orthogonal imaginary units and obey Hamilton's multiplication rules

$$\mathbf{i}^2 = \mathbf{j}^2 = \mathbf{k}^2 = -1, \mathbf{ij} = -\mathbf{ji} = \mathbf{k}, \mathbf{jk} = -\mathbf{kj} = \mathbf{i}, \mathbf{ki} = -\mathbf{ik} = \mathbf{j} \quad (7)$$

For a quaternion number $q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} \in \mathbb{H}$, q_0 is called as the scalar part of q and denoted by $Sc(q)$. $q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$ is named as the vector (or pure) part of q and conventionally notated by $\mathbf{q}$. The conjugate $\bar{q}$ of a quaternion q is given by

$$\bar{q} = q_0 - q_1\mathbf{i} - q_2\mathbf{j} - q_3\mathbf{k} \quad (8)$$

which is an anti-involution, i.e.,

$$\overline{qp} = \bar{p}\bar{q} \quad (9)$$

According to Eq. (8), the norm or modulus of $q \in \mathbb{H}$ is defined by

$$|q| = \sqrt{q\bar{q}} = \sqrt{q_0^2 + q_1^2 + q_2^2 + q_3^2} \quad (10)$$

and the inverse of $q \in \mathbb{H} \setminus \{0\}$ is

$$q^{-1} = \frac{\bar{q}}{|q|^2} \quad (11)$$

Any quaternion q may be split up into [21]

$$q = q_+ + q_-, \quad q_{\pm} = \frac{1}{2}(q \pm \mathbf{i}q\mathbf{j}) \quad (12)$$

This leads to the following modulus identity

$$|q|^2 = |q_+|^2 + |q_-|^2 \quad (13)$$

A quaternion-valued function $f: \mathbb{R}^2 \rightarrow \mathbb{H}$ can be expressed as

$$f(\mathbf{x}) = f_0(\mathbf{x}) + f_1(\mathbf{x})\mathbf{i} + f_2(\mathbf{x})\mathbf{j} + f_3(\mathbf{x})\mathbf{k}, \quad f_0(\mathbf{x}), f_1(\mathbf{x}), f_2(\mathbf{x}), f_3(\mathbf{x}) \in \mathbb{R} \quad (14)$$

The connection between $\mathbb{H}$ and Clifford algebra is presented clearly in literature [6].

2.2. The Quaternion Linear Canonical Transform

Definition 1. Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. The two-sided QLCT of $f \in L^1(\mathbb{R}^2, \mathbb{H})$ is defined by [22, 23]

$$\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u}) := \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1, u_1)f(\mathbf{x})K_{A_2}^{\mathbf{j}}(x_2, u_2)d\mathbf{x} \quad (15)$$

where the kernel functions of the QLCT above are given by

$$K_{A_1}^{\mathbf{i}}(x_1, u_1) := \begin{cases} \frac{1}{\sqrt{2\pi b_1}} e^{\mathbf{i}\left(\frac{a_1}{2b_1}x_1^2 - \frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1}u_1^2 - \frac{\pi}{4}\right)}, & b_1 \neq 0 \\ \sqrt{d_1} e^{\mathbf{i}\frac{c_1 d_1}{2}u_1^2}, & b_1 = 0 \end{cases} \quad (16)$$

and

$$K_{A_2}^{\mathbf{j}}(x_2, u_2) := \begin{cases} \frac{1}{\sqrt{2\pi b_2}} e^{\mathbf{j}\left(\frac{a_2}{2b_2}x_2^2 - \frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2}u_2^2 - \frac{\pi}{4}\right)}, & b_2 \neq 0 \\ \sqrt{d_2} e^{\mathbf{j}\frac{c_2 d_2}{2}u_2^2}, & b_2 = 0 \end{cases} \quad (17)$$

Here, $\mathbf{x} = x_1\mathbf{e}_1 + x_2\mathbf{e}_2$ and $\mathbf{u} = u_1\mathbf{e}_1 + u_2\mathbf{e}_2$. From the above definition, it is noted that for $b_i = 0, i = 1, 2$ the QLCT of a signal is essentially a chirp multiplication and it is of no particular interest for our objective in this work. Hence, without loss of generality, we set $b_i \neq 0$ in the following section unless stated otherwise. Under some suitable conditions, the QLCT above is invertible and the inversion of this transform is given in Definition 2.

Definition 2. Suppose $f \in L^1(\mathbb{R}^2, \mathbb{H})$, then the inversion of the two-sided QLCT of f is given by [22, 23]

$$\begin{aligned} f(\mathbf{x}) &:= \{\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\}^{-1}\{f\}(\mathbf{x}) \\ &:= \int_{\mathbb{R}^2} K_{A_1}^{-\mathbf{i}}(x_1, u_1)\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u})K_{A_2}^{-\mathbf{j}}(x_2, u_2)d\mathbf{u} \end{aligned} \quad (18)$$

Many other important properties of the QLCT have been derived [22, 23] and an application of the QLCT to study the generalized swept-frequency filters was presented in [3]. To be attention, in this paper, we mainly investigate UP on the two-sided QLCT. Without explanation, $\mathcal{F}$ denotes the two-sided QFT operator and $\mathcal{L}_{A_1, A_2}^{\mathbb{H}}$ is the two-sided QLCT operator.

2.3. Relationship Between the QLCT and QFT

Recently, the fundamental relationship between the QLCT and QFT is derived [5] and it provides a simple proof to obtain most of important properties of QLCT such as inversion property, Parseval formula and time-frequency shift properties. It plays a key role and greatly simplifies to verify the Heisenberg type uncertainty principle on the QLCT. The QLCT can be reduced to QFT as follows

$$\mathcal{F}\{g_f\}(\mathbf{u}) = \frac{1}{\sqrt{(2\pi)^2}} \int_{\mathbb{R}^2} e^{-i\mathbf{x}_1 u_1} g_f(\mathbf{x}) e^{-j\mathbf{x}_2 u_2} d\mathbf{x} \quad (19)$$

where

$$\begin{aligned} \mathcal{F}\{g_f\}(\mathbf{u}) &= \tilde{\mathcal{F}}(\mathbf{b}\mathbf{u}) \\ g_f(\mathbf{x}) &= \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} \tilde{f}(\mathbf{x}) \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \\ \tilde{f}(\mathbf{x}) &= e^{i(a_1/2b_1)x_1^2} f(\mathbf{x}) e^{j(a_2/2b_2)x_2^2} \end{aligned} \quad (20)$$

with

$$\begin{aligned} \tilde{\mathcal{F}}(\mathbf{u}) &= \frac{1}{\sqrt{(2\pi)^2}} \int_{\mathbb{R}^2} e^{-i\mathbf{x}_1(u_1/b_1)} g_f(\mathbf{x}) e^{-j\mathbf{x}_2(u_2/b_2)} d\mathbf{x} \\ &= e^{-i(d_1/2b_1)u_1^2} \mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u}) e^{-j(d_2/2b_2)u_2^2} \end{aligned} \quad (21)$$

Herein, we put $\mathbf{b} = (b_1, b_2) \in \mathbb{R}^2$, the element wise multiplication $\mathbf{b}\mathbf{u} = (b_1 u_1, b_2 u_2)$, and the element wise division $\frac{\mathbf{u}}{\mathbf{b}} = \left(\frac{u_1}{b_1}, \frac{u_2}{b_2}\right)$. The specific proof of this relationship is in [5].

3. Uncertainty Principles for the QLCT

The main results of this paper are presented in this section.

Theorem 1. *Let $f \in L^1(\mathbb{R}^2, \mathbb{H}) \cap L^2(\mathbb{R}^2, \mathbb{H})$ be a quaternion-valued signal, then the directional uncertainty principle associated with the QLCT is given by as follows*

$$\begin{aligned} &\int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |f(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} (\boldsymbol{\gamma} \cdot \mathbf{u})^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u})|^2 d\mathbf{u} \\ &\geq \frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \mathbf{b}\boldsymbol{\gamma})^2 F_-^2 + (\boldsymbol{\alpha} \cdot \mathbf{b}\boldsymbol{\gamma}')^2 F_+^2 \right] \end{aligned} \quad (22)$$

where constant vectors $\boldsymbol{\alpha}, \boldsymbol{\gamma} \in \mathbb{R}^2$ and $\boldsymbol{\gamma}' = -\gamma_1 \mathbf{e}_1 + \gamma_2 \mathbf{e}_2$. Besides, $F_- = \int_{\mathbb{R}^2} |f_-(\mathbf{x})|^2 d\mathbf{x}$, $F_+ = \int_{\mathbb{R}^2} |f_+(\mathbf{x})|^2 d\mathbf{x}$.

Proof. Substituting the quaternion function f by g_f defined in Eq. (20) in both sides of Eq. (4), it follows that

$$\begin{aligned} &\int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |g_f(\mathbf{x})|^2 d\mathbf{x} \int_{\mathbb{R}^2} (\boldsymbol{\beta} \cdot \mathbf{u})^2 |\mathcal{F}\{g_f\}(\mathbf{u})|^2 d\mathbf{u} \\ &\geq \frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \boldsymbol{\beta})^2 G_{f-}^2 + (\boldsymbol{\alpha} \cdot \boldsymbol{\beta}')^2 G_{f+}^2 \right] \end{aligned} \quad (23)$$

Setting $\mathbf{u} = \mathbf{b}\mathbf{u}$, the left side of Eq. (4) further becomes

$$\begin{aligned}
 & \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 \left| \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} \tilde{f}(\mathbf{x}) \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \right|^2 d\mathbf{x} \cdot \frac{1}{\mathbf{b}} \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 |\tilde{\mathcal{F}}(\mathbf{u})|^2 d\mathbf{u} \\
 &= \frac{1}{b_1 b_2} \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |\tilde{f}(\mathbf{x})|^2 d\mathbf{x} \cdot \frac{1}{\mathbf{b}} \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 \\
 & \quad \left| e^{-i(d_1/2b_1)u_1^2} \mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u}) e^{-j(d_2/2b_2)u_2^2} \right|^2 d\mathbf{u} \\
 &= \frac{1}{(b_1 b_2)^2} \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |\tilde{f}(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \\
 &= \frac{1}{(b_1 b_2)^2} \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 \left| e^{i(a_1/2b_1)x_1^2} f(\mathbf{x}) e^{j(a_2/2b_2)x_2^2} \right|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 \\
 & \quad |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \\
 &= \frac{1}{(b_1 b_2)^2} \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |f(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \quad (24)
 \end{aligned}$$

It is easy to verify that

$$\begin{aligned}
 G_{f_-} &= \int_{\mathbb{R}^2} |g_{f_-}(\mathbf{x})|^2 d\mathbf{x} \\
 &= \int_{\mathbb{R}^2} \left| \frac{1}{\sqrt{b_1}} e^{i(\frac{a_1}{2b_1}x_1^2 - \frac{\pi}{4})} f_-(\mathbf{x}) e^{j(\frac{a_2}{2b_2}x_2^2 - \frac{\pi}{4})} \frac{1}{\sqrt{b_2}} \right|^2 d\mathbf{x} \\
 &= \frac{1}{b_1 b_2} \int_{\mathbb{R}^2} |f_-(\mathbf{x})|^2 d\mathbf{x} = \frac{1}{b_1 b_2} F_- \quad (25)
 \end{aligned}$$

and $G_{f_+} = \frac{1}{b_1 b_2} \int_{\mathbb{R}^2} |f_+(\mathbf{x})|^2 d\mathbf{x} = \frac{1}{b_1 b_2} F_+$. Thus, the right side of Eq. (23) is

$$\frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \boldsymbol{\beta})^2 G_{f_-}^2 + (\boldsymbol{\alpha} \cdot \boldsymbol{\beta}')^2 G_{f_+}^2 \right] = \frac{1}{4(b_1 b_2)^2} \left[(\boldsymbol{\alpha} \cdot \boldsymbol{\beta})^2 F_-^2 + (\boldsymbol{\alpha} \cdot \boldsymbol{\beta}')^2 F_+^2 \right] \quad (26)$$

We derive that

$$\begin{aligned}
 & \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |f(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} \left(\boldsymbol{\beta} \cdot \frac{\mathbf{u}}{\mathbf{b}} \right)^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \\
 & \geq \frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \boldsymbol{\beta})^2 F_-^2 + (\boldsymbol{\alpha} \cdot \boldsymbol{\beta}')^2 F_+^2 \right] \quad (27)
 \end{aligned}$$

Substituting $\boldsymbol{\gamma} = \frac{\boldsymbol{\beta}}{\mathbf{b}}$, we have

$$\begin{aligned}
 & \int_{\mathbb{R}^2} (\boldsymbol{\alpha} \cdot \mathbf{x})^2 |f(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} (\boldsymbol{\gamma} \cdot \mathbf{u})^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \\
 & \geq \frac{1}{4} \left[(\boldsymbol{\alpha} \cdot \mathbf{b}\boldsymbol{\gamma})^2 F_-^2 + (\boldsymbol{\alpha} \cdot \mathbf{b}\boldsymbol{\gamma}')^2 F_+^2 \right] \quad (28)
 \end{aligned}$$

This finishes the proof of the theorem. $\square$

Remark 1. It is not difficult to show that the UP in Theorem 1 generalizes other UPs in previously developed uncertainty relations. The UP derived in Eq.(4) can be shown as a special case for the matrix parameter of the QLCT

$$A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}.$$

Remark 2. Now choosing $\alpha = \gamma = \mathbf{e}_k, k = 1, 2$, the UP in Eq. (22) becomes

$$\begin{aligned} & \int_{\mathbb{R}^2} x_k^2 |f(\mathbf{x})|^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} u_k^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 d\mathbf{u} \\ & \geq \frac{1}{4} [b_k^2 F_-^2 + b_k^2 F_+^2] = \frac{b_k^2}{4} F^2 \end{aligned} \quad (29)$$

This shows that the directional UP in Theorem 1 is a generalization of the component-wise UP for the QLCT derived in [24].

However, the lower bound of these principles is quite similar to that of the Heisenberg UP for the 2D FT and LCT [4]. In the following theorem, we give a modified UP for the two-sided QLCT, i.e., full uncertainty principle for the QLCT.

Theorem 2. Let $f \in L^1(\mathbb{R}^2, \mathbb{H}) \cap L^2(\mathbb{R}^2, \mathbb{H})$ be a quaternion-valued signal, a modified UP is given by

$$\begin{aligned} & \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} u_1^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}} \{f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u} \\ & \geq (b_1 b_2)^2 \left| \frac{1}{2} F - \int_{\mathbb{R}^2} \left| x_1 J_{A_1, A_2}^f(\mathbf{x}) x_2 \right| d\mathbf{x} \right|^2 \end{aligned} \quad (30)$$

where

$$\begin{aligned} J_{A_1, A_2}^f(\mathbf{x}) &= \frac{a_1 x_1}{b_1} \frac{\overline{\partial f(\mathbf{x})}}{\partial x_2} \mathbf{i} f(\mathbf{x}) + \frac{\overline{\partial f(\mathbf{x})}}{\partial x_2} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ &\quad - \frac{a_2 x_2}{b_2} \mathbf{j} \overline{f(\mathbf{x})} \mathbf{i} f(\mathbf{x}) \frac{a_1 x_1}{b_1} - \frac{a_2 x_2}{b_2} \mathbf{j} \overline{f(\mathbf{x})} \frac{\partial f(\mathbf{x})}{\partial x_1} \end{aligned} \quad (31)$$

and $F = \int_{\mathbb{R}^2} |f(\mathbf{x})|^2 d\mathbf{x}$.

Proof. Similar to the proof of Theorem 1, we firstly replace f with g_f in Eq. (5) and derive

$$\begin{aligned} & \int_{\mathbb{R}^2} x_1^2 |g_f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} u_1^2 |\mathcal{F}\{g_f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u} \\ & \geq \left| \frac{1}{2} G_f - \int_{\mathbb{R}^2} \left| x_1 x_2 \frac{\overline{\partial g_f(\mathbf{x})}}{\partial x_2} \frac{\partial g_f(\mathbf{x})}{\partial x_1} \right| d\mathbf{x} \right|^2 \end{aligned} \quad (32)$$

Based on Eq. (20), it is easy to obtain

$$g_f(\mathbf{x}) = \frac{e^{-\mathbf{i}(\pi/4)}}{\sqrt{b_1}} e^{\mathbf{i}(a_1/2b_1)x_1^2} f(\mathbf{x}) e^{\mathbf{j}(a_2/2b_2)x_2^2} \frac{e^{-\mathbf{j}(\pi/4)}}{\sqrt{b_2}} \quad (33)$$

and $G_f = \int_{\mathbb{R}^2} |g_f(\mathbf{x})|^2 d\mathbf{x} = \frac{1}{b_1 b_2} \int_{\mathbb{R}^2} |f(\mathbf{x})|^2 d\mathbf{x} = \frac{1}{b_1 b_2} F$.

Then,

$$\begin{aligned}
 \frac{\partial g_f(\mathbf{x})}{\partial x_1} &= \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} \frac{\partial}{\partial x_1} \left(e^{i(a_1/2b_1)x_1^2} f(\mathbf{x}) \right) e^{j(a_2/2b_2)x_2^2} \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \\
 &= \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} e^{i(a_1/2b_1)x_1^2} \left(i \frac{a_1 x_1}{b_1} f(\mathbf{x}) + \frac{\partial f(\mathbf{x})}{\partial x_1} \right) e^{j(a_2/2b_2)x_2^2} \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \\
 \frac{\partial g_f(\mathbf{x})}{\partial x_2} &= \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} e^{i(a_1/2b_1)x_1^2} \frac{\partial}{\partial x_2} \left(f(\mathbf{x}) e^{j(a_2/2b_2)x_2^2} \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \right) \\
 &= \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} e^{i(a_1/2b_1)x_1^2} \left(\frac{\partial f(\mathbf{x})}{\partial x_2} + f(\mathbf{x}) j \frac{a_2 x_2}{b_2} \right) e^{j(a_2/2b_2)x_2^2} \frac{e^{-j(\pi/4)}}{\sqrt{b_2}}
 \end{aligned} \tag{34}$$

By direct calculation, we derive

$$\begin{aligned}
 \frac{\partial g_f(\mathbf{x})}{\partial x_2} \frac{\partial g_f(\mathbf{x})}{\partial x_1} &= \frac{e^{j(\pi/4)}}{\sqrt{b_2}} e^{-j(a_2/2b_2)x_2^2} \left(\frac{\partial f(\mathbf{x})}{\partial x_2} - j f(\mathbf{x}) \frac{a_2 x_2}{b_2} \right) \\
 &\quad e^{-i(a_1/2b_1)x_1^2} \frac{e^{i(\pi/4)}}{\sqrt{b_1}} \cdot \frac{e^{-i(\pi/4)}}{\sqrt{b_1}} e^{i(a_1/2b_1)x_1^2} \\
 &\quad \times \left(i \frac{a_1 x_1}{b_1} f(\mathbf{x}) + \frac{\partial f(\mathbf{x})}{\partial x_1} \right) e^{j(a_2/2b_2)x_2^2} \frac{e^{-j(\pi/4)}}{\sqrt{b_2}} \\
 &= \frac{1}{b_1 b_2} e^{j(\pi/4)} e^{-j(a_2/2b_2)x_2^2} J_{A_1, A_2}^f(\mathbf{x}) e^{j(a_2/2b_2)x_2^2} e^{-j(\pi/4)}
 \end{aligned}$$

where $J_{A_1, A_2}^f(\mathbf{x}) = \frac{a_1 x_1}{b_1} \frac{\partial f(\mathbf{x})}{\partial x_2} i f(\mathbf{x}) + \frac{\partial f(\mathbf{x})}{\partial x_2} \frac{\partial f(\mathbf{x})}{\partial x_1} - \frac{a_2 x_2}{b_2} j f(\mathbf{x}) i f(\mathbf{x})$
 $\frac{a_1 x_1}{b_1} - \frac{a_2 x_2}{b_2} j f(\mathbf{x}) \frac{\partial f(\mathbf{x})}{\partial x_1}$.

Setting $\mathbf{u} = \mathbf{b}\mathbf{u}$, the left side of Eq. (32) changes into

$$\begin{aligned}
 \int_{\mathbb{R}^2} x_1^2 |g_f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \int_{\mathbb{R}^2} u_1^2 |\mathcal{F}\{g_f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u} &= \frac{1}{b_1 b_2} \int_{\mathbb{R}^2} x_1^2 \\
 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \cdot \frac{1}{b} \int_{\mathbb{R}^2} \frac{u_1^2}{b_1^2} \left| e^{-i(d_1/2b_1)u_1^2} \mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u}) e^{-j(d_2/2b_2)u_2^2} \right|^2 \frac{u_2^2}{b_2^2} d\mathbf{u} \\
 &= \frac{1}{(b_1 b_2)^4} \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} u_1^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u}
 \end{aligned} \tag{35}$$

Above all, we derive

$$\begin{aligned}
 \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \cdot \int_{\mathbb{R}^2} u_1^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u} &\geq (b_1 b_2)^4 \\
 \left| \frac{1}{2} \frac{1}{b_1 b_2} F - \int_{\mathbb{R}^2} \left| x_1 x_2 \frac{1}{b_1 b_2} e^{j(\pi/4)} e^{-j(a_2/2b_2)x_2^2} J_{A_1, A_2}^f(\mathbf{x}) e^{j(a_2/2b_2)x_2^2} e^{-j(\pi/4)} \right| d\mathbf{x} \right|^2 \\
 &= (b_1 b_2)^2 \left| \frac{1}{2} F - \int_{\mathbb{R}^2} |x_1 x_2 J_{A_1, A_2}^f(\mathbf{x})| d\mathbf{x} \right|^2
 \end{aligned} \tag{36}$$

The proof is complete. $\square$

Remark 3. We can draw the conclusion that the UP in Theorem 2 is an extended result based on the QLCT. Similar to the Remark 1, when $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$, Eq. (30) is the modified UP for the QFT in Eq. (5).

Remark 4. Common proofs in [4] demonstrate that the equality in Eq. (5) holds and the lowest bound is derived for a 2D Gaussian function $g_f(\mathbf{x}) = K_0 e^{-\left(\frac{x_1^2}{2\alpha_1} + \frac{x_2^2}{2\alpha_2}\right)}$ with that $\alpha_1 > 0, \alpha_2 > 0$ are real constants and K_0 is a quaternion constant. Thus, based on Eq. (33), we conclude that the equality in Eq. (30) holds when

$$f(\mathbf{x}) = \sqrt{b_1} e^{\mathbf{j}\frac{\pi}{4}} e^{-\mathbf{i}\frac{a_1}{2b_1}x_1^2} K_0 e^{-\left(\frac{x_1^2}{2\alpha_1} + \frac{x_2^2}{2\alpha_2}\right)} e^{-\mathbf{j}\frac{a_2}{2b_2}x_2^2} e^{\mathbf{j}\frac{\pi}{4}} \sqrt{b_2} \quad (37)$$

In the following context, an example is given to verify this result.

Example 1. Consider a two-dimensional quaternion function of the form

$$f(\mathbf{x}) = \frac{1}{(\pi^2 \alpha_1 \alpha_2)^{1/4}} \sqrt{b_1} e^{\mathbf{j}\frac{\pi}{4}} e^{-\mathbf{i}\frac{a_1}{2b_1}x_1^2} e^{-\left(\frac{x_1^2}{2\alpha_1} + \frac{x_2^2}{2\alpha_2}\right)} e^{-\mathbf{j}\frac{a_2}{2b_2}x_2^2} e^{\mathbf{j}\frac{\pi}{4}} \sqrt{b_2} \quad (38)$$

It follows that

$$F = \int_{\mathbb{R}^2} |f(\mathbf{x})|^2 d\mathbf{x} = \frac{b_1 b_2}{(\pi^2 \alpha_1 \alpha_2)^{1/2}} \int_{\mathbb{R}^2} e^{-\left(\frac{x_1^2}{\alpha_1} + \frac{x_2^2}{\alpha_2}\right)} d\mathbf{x} = b_1 b_2 \quad (39)$$

The QLCT of $f(\mathbf{x})$ is

$$\begin{aligned} \mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u}) &= \int_{\mathbb{R}^2} K_{A_1}^{\mathbf{i}}(x_1, u_1) f(\mathbf{x}) K_{A_2}^{\mathbf{j}}(x_2, u_2) d\mathbf{x} \\ &= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}^2} e^{\mathbf{i}\left(-\frac{x_1 u_1}{b_1} + \frac{d_1}{2b_1} u_1^2\right)} \frac{1}{(\pi^2 \alpha_1 \alpha_2)^{1/4}} e^{-\left(\frac{x_1^2}{2\alpha_1} + \frac{x_2^2}{2\alpha_2}\right)} \frac{1}{\sqrt{2\pi}} e^{\mathbf{i}\left(-\frac{x_2 u_2}{b_2} + \frac{d_2}{2b_2} u_2^2\right)} d\mathbf{x} \\ &= \frac{1}{2\pi} \frac{1}{(\pi^2 \alpha_1 \alpha_2)^{1/4}} e^{\mathbf{i}\frac{d_1}{2b_1} u_1^2} \int_{\mathbb{R}^2} e^{-\mathbf{i}\frac{u_1}{b_1} x_1 - \frac{x_1^2}{2\alpha_1} - \frac{x_2^2}{2\alpha_2} - \mathbf{j}\frac{u_2}{b_2} x_2} d\mathbf{x} \cdot e^{\mathbf{j}\frac{d_2}{2b_2} u_2^2} \\ &= \frac{(\alpha_1 \alpha_2)^{1/4}}{\sqrt{\pi}} e^{\mathbf{i}\frac{d_1}{2b_1} u_1^2 - \frac{\alpha_1}{2b_1^2} u_1^2} \cdot e^{\mathbf{j}\frac{d_2}{2b_2} u_2^2 - \frac{\alpha_2}{2b_2^2} u_2^2} \end{aligned} \quad (40)$$

Further, we have

$$\begin{aligned} \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} &= \frac{b_1 b_2}{(\pi^2 \alpha_1 \alpha_2)^{1/2}} \int_{\mathbb{R}^2} x_1^2 e^{-\frac{x_1^2}{\alpha_1} - \frac{x_2^2}{\alpha_2}} x_2^2 d\mathbf{x} \\ &= \frac{b_1 b_2}{4(\pi^2 \alpha_1 \alpha_2)^{1/2}} \frac{\sqrt{\pi}}{\alpha_1^{-3/2}} \frac{\sqrt{\pi}}{\alpha_2^{-3/2}} = \frac{b_1 b_2 (\alpha_1 \alpha_2)}{4} \end{aligned} \quad (41)$$

Besides,

$$\begin{aligned} \int_{\mathbb{R}^2} u_1^2 \left| \mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u}) \right|^2 u_2^2 d\mathbf{u} &= \frac{(\alpha_1 \alpha_2)^{1/2}}{\pi} \int_{\mathbb{R}^2} u_1^2 e^{-\frac{\alpha_1}{b_1^2} u_1^2 - \frac{\alpha_2}{b_2^2} u_2^2} u_2^2 d\mathbf{u} \\ &= \frac{(\alpha_1 \alpha_2)^{1/2}}{4\pi} \frac{\sqrt{\pi}}{(b_1^2/\alpha_1)^{-3/2}} \frac{\sqrt{\pi}}{(b_2^2/\alpha_2)^{-3/2}} = \frac{(b_1 b_2)^3}{4(\alpha_1 \alpha_2)} \end{aligned} \quad (42)$$

It means

$$\begin{aligned} \int_{\mathbb{R}^2} x_1^2 |f(\mathbf{x})|^2 x_2^2 d\mathbf{x} \int_{\mathbb{R}^2} u_1^2 |\mathcal{L}_{A_1, A_2}^{\mathbb{H}}\{f\}(\mathbf{u})|^2 u_2^2 d\mathbf{u} &= \frac{b_1 b_2 (\alpha_1 \alpha_2)}{4} \frac{(b_1 b_2)^3}{4 (\alpha_1 \alpha_2)} \\ &= \frac{(b_1 b_2)^4}{16} \end{aligned} \quad (43)$$

4. Potential Applications

Two novel UPs in the QLCT setting have been derived in this paper. It can be regarded as the first step in the applications of the QLCT in the real world. In this subsection, the potential applications of derived results in the QLCT domain are therefore proposed to show the usefulness and effectiveness of the theory.

UPs have many practical applications. For example, UPs of the LCT can be used for the analysis of optical systems related to the spread in the LCT domain [32]. As for a generalization of transformation from complex to quaternion algebra, the QLCT transforms a quaternionic 2D signal into a quaternion-valued frequency domain signal, which is an effective processing tool for color image analysis. The UPs of the QLCT can state that the effective widths of quaternion-valued signal in spatial and frequency domains. Thus, the derived Theorems 1 and 2 can be used in the estimation of effective bandwidth in the QLCT domain. The condition of equality can offer special wave forms to provide multi-scale spectral resolution. Besides, Theorem 1 can further estimate the bandwidth in arbitrary but fixed directions.

At the same time, an analog signal is usually represented by its discrete samples in practice. Discrete UPs are very useful and closely relate to the problem of signal recovery [13, 25, 30]. An iterative method for signal recovery problems has been proposed by Kou et.al in quaternion analysis setting, i.e., the QFT [25]. Compared with the QFT, the QLCT is more flexible and suitable for the process of non-stationary signals. Then, newly derived UPs are able to be extended into discrete version and related to the signal recovery problems in the sense of the QLCT. Therefore, the UPs of Eqs. (4) and (5) in the QLCT domain presented in this paper are interesting and useful. Real applications of the derived results are also our future research directions.

5. Conclusions

In this paper, using the relationship between the QFT and QLCT, we obtain two novel UPs for the two-sided QLCT and find the lower bound on the product of the spreads of quaternion-valued function and its QLCT. Firstly, the directional UP for the QLCT is given in Theorem 1 and the lower bound is different from the QFT, which is tightly close to the parameters of QLCT and selecting directions. It can be regarded as a generalization of component-wise uncertainty principle on QLCT as well. Secondly, a modified UP, also called as full UP, is derived for the QLCT in Theorem 2. The lower bound is not only related to parameters of the QLCT, but also the characteristics

of the signal itself. Both theorems are generalizations of common UP in the QFT. Finally, an example is given to derive the lowest bound and verify the correctness of the result. Further investigations and extensions of this work will focus on the other two QLCTs, different types of UP on quaternion integral transforms, and discrete UPs in the QLCT setting.

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